

Problem Statement

Project Title

AI-Powered Personalized Nutrition Agent using IBM watsonx Orchestrate

Problem Statement

Many people struggle to maintain a healthy diet because they do not know how to create a balanced meal plan based on their age, weight, health conditions, food preferences, and lifestyle. Generic diet plans often ignore individual needs, leading to poor nutrition and health issues.

There is a need for an intelligent assistant that can provide personalized nutrition guidance by considering user-specific information and trusted dietary knowledge.

Objectives

- Develop an AI-powered Nutrition Agent using IBM watsonx Orchestrate.
 - Collect user information such as age, gender, weight, height, activity level, food preferences, and medical conditions.
 - Generate personalized meal and nutrition plans.
 - Provide healthy lifestyle recommendations.
 - Answer nutrition-related questions using a knowledge base.
 - Promote healthy eating habits and improve overall wellness.
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Proposed Solution

The proposed solution is an AI Nutrition Agent developed using IBM watsonx Orchestrate. The agent interacts with users through natural language, gathers their health and dietary information, and provides personalized diet recommendations.

The agent uses a knowledge base containing nutrition guidelines, BMI information, healthy lifestyle recommendations, and Indian meal plans to deliver accurate and relevant responses.

IBM Technologies Used

- IBM Cloud

- IBM watsonx Orchestrate
 - IBM Foundation Models (GPT-OSS 120B)
 - Knowledge Base
 - AI Agent Builder
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Expected Outcome

The developed Nutrition Agent will:

- Recommend personalized diet plans.
 - Suggest healthy lifestyle improvements.
 - Answer nutrition-related queries accurately.
 - Promote balanced eating habits.
 - Provide a user-friendly AI-powered nutrition assistant.
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Conclusion

This project demonstrates how IBM watsonx Orchestrate can be used to build intelligent AI agents that provide personalized nutrition guidance. By combining foundation models with a curated knowledge base, the agent delivers practical, user-specific recommendations and showcases the potential of AI in healthcare and wellness.